Wei Wang (王尉)

Ph.D. Candidate in Mathematics, School of Mathematical Sciences, Peking University

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Education

Peking University, School of Mathematical Sciences *Beijing, China*

Ph.D. Candidate in Mathematics Sept. 2022 – 2027 (expected)

Advisor: Prof. Zhifei Zhang

Zhejiang University, School of Mathematical Sciences *Hangzhou, Zhejiang, China*

B.S. in Mathematics and Applied Mathematics Sept. 2018 – June 2022

Advisor: Prof. Ting Zhang

Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

Research Interests

Geometric variational problems; elliptic and parabolic partial differential equations; harmonic maps and their flows; liquid crystal theory; homogenization theory; nonlocal operators.

Publications and Accepted Papers

(8) Sharp convergence rates for parabolic Green's functions in time-independent periodic homogenization.
Wei Wang.
Accepted for publication in *Frontiers of Mathematics*. [arXiv: 2606.12101]

(7) Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers.
Wei Wang and Zhifei Zhang.
Accepted for publication in *Memoirs of the European Mathematical Society*. [arXiv: 2404.00677]

(6) Existence of weak solutions for two-phase matrix-valued harmonic map flows.
Wei Wang, Wei Wang (王伟), and Zhifei Zhang.
Journal of Functional Analysis 291 (2026), no. 11, Article No. 111623. [doi] [arXiv: 2506.16278]

(5) Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects.
Haotong Fu, Huaijie Wang, and **Wei Wang**.
SIAM Journal on Mathematical Analysis 58 (2026), no. 4, 3306–3324. [doi] [arXiv: 2508.01811]

(4) Stratification and rectifiability of harmonic map flows via tangent measures.
Haotong Fu, **Wei Wang**, Ke Wu, and Zhifei Zhang.
Mathematische Annalen 395 (2026), Article No. 92. [doi] [arXiv: 2504.14880]

(3) On the sharp constants in the regional fractional Sobolev inequalities.
Rupert L. Frank, Tianling Jin, and **Wei Wang**.
SN Partial Differential Equations and Applications 6 (2025), Article No. 15. [doi] [arXiv: 2403.00357]

(2) Uniform estimates of resolvents in homogenization theory of elliptic systems.
Wei Wang.
Journal of Differential Equations 370 (2023), 1–65. [doi] [arXiv: 2211.14614]

(1) Homogenization theory of elliptic systems with lower order terms in dimension two.
Wei Wang and Ting Zhang.
Communications on Pure and Applied Analysis 22 (2023), no. 3, 787–824. [doi] [arXiv: 2210.06329]

Preprints and Manuscripts

(13) Asymptotics for two-dimensional parabolic vectorial Allen–Cahn systems.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.
Preprint. [arXiv: 2609.27596]

(12) Small-energy regularity for stationary weak solutions of the Liouville equation.
Haotong Fu, Kelei Wang, **Wei Wang**, and Ke Wu.
Preprint. [arXiv: 2609.26026]

(11) Boundary asymptotics for two-dimensional vectorial Allen–Cahn systems.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.
Preprint. [arXiv: 2609.00739]

(10) Boundary regularity for Oseen–Frank minimizers with arbitrary positive splay, twist, and bend constants.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**.
Preprint. [arXiv: 2608.22287]

(9) Sharp instability of planar isotropic–nematic interfaces in the Landau–de Gennes model.
Wei Wang and Qin Wu.
Preprint. [arXiv: 2608.12131]

(8) Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime.
Haotong Fu, Huaijie Wang, and **Wei Wang**.
Preprint. [arXiv: 2608.08456]

(7) On the absence of point defects in biaxial Landau–de Gennes models.
Haotong Fu, Huaijie Wang, **Wei Wang**, and Zhifei Zhang.
Preprint. [arXiv: 2608.04908]

(6) Singular limits for three-dimensional global minimizers of Ginzburg–Landau-type functionals: uniform estimates and singular sets.
Giacomo Canevari, Haotong Fu, and **Wei Wang**.
Preprint. [arXiv: 2606.27691]

(5) A note on an approximate fixed point theorem for VMO maps.
Haotong Fu and **Wei Wang**.
Preprint. [arXiv: 2606.15670]

(4) Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions.
Giacomo Canevari, Haotong Fu, and **Wei Wang**.
Preprint. [arXiv: 2605.04442]

(3) Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit.
Haotong Fu, Huaijie Wang, and **Wei Wang**.
Preprint. [arXiv: 2507.14955]

(2) Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity.
Wei Wang and Zhifei Zhang.
Preprint. [arXiv: 2411.16048]

(1) Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations.
Haotong Fu, **Wei Wang**, and Zhifei Zhang.
Preprint. [arXiv: 2408.06726]

Research Visits and Programs

University of Verona, Department of Computer Science *Verona, Italy*

Visiting Ph.D. Student Mar. – Apr. 2026

Host: Prof. Giacomo Canevari

HKUST, Department of Mathematics *Hong Kong, China*

Summer Research Program June – Sept. 2021

Advisor: Prof. Tianling Jin

Selected Talks

- Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals: Uniform Estimates and Singular Sets
  Conference on Geometric Measure Theory and Partial Differential Equations
  Enshi, Hubei, China, Aug. 2026.
- Uniform Estimates for Landau–de Gennes Minimizers with Point and Line Defects
  Academic Forum on Theory and Numerical Analysis of Liquid Crystals and Complex Systems
  Xiangtan University, July 2026.
- Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers
  Academic Workshop on Fluid Dynamics Equations
  Peking University, Jan. 2026.
- Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity
  Jinhua Conference on Harmonic Analysis and Its Applications
  Zhejiang Normal University, Nov. 2025.
- Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit
  Conference on Theory and Computation of Quasi Crystals
  Nankai University, Aug. 2025.
- Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures
  PDE and Geometry Seminar
  Zhejiang University, June 2025.

Grants and Projects

- **Physical Modelling and Mathematical Theory of Liquid Crystals** 2023 – 2028
  Participant, NSFC, Grant No. 12288101.
- **Physical Modeling and Mathematical Theory of Liquid Crystal Systems** 2023 – 2028
  Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

Academic Service

Mathematical Reviews / MathSciNet, American Mathematical Society

Registered prospective reviewer.

Teaching Experience

- **Peking University, Teaching Assistant** 2022 – 2026
  Advanced Mathematics (A, B, and C); Real Analysis; Functional Analysis (I); Harmonic Analysis.
- **Zhejiang University, Teaching Assistant** 2021 – 2022
  Mathematical Analysis (I and II).

Honors and Awards

- National Scholarship Sept. 2026
  School of Mathematical Sciences, Peking University
- First Prize, Graduate Research Exhibition Apr. 2025
  School of Mathematical Sciences, Peking University
- Third Prize, Graduate Research Exhibition Apr. 2024
  School of Mathematical Sciences, Peking University
- Elite Program for Graduate Students in Mathematics Oct. 2023
  Peking University
- Nomination Award, ICCM Creative Undergraduate Thesis Aug. 2022
  Shing-Tung Yau Center of Southeast University
- First Prize, 11th Chinese Mathematics Competitions Finals Apr. 2021
  Wuhan University
- Nomination Award, S.-T. Yau College Student Mathematics Contest June 2021
  Yau Mathematical Sciences Center, Tsinghua University

Skills: LaTeX, MATLAB, Mathematica, Python, Lean. **Languages:** Chinese (native), English (fluent).